

Chapter 12: Ratios, rates and time

11 lessons · one lesson for each exercise · each lesson is one hour

NSW syllabus outcome

Stage 4 Number and algebra: MA4-RAT-C-01 solves problems involving ratios and rates, and analyses distance–time graphs; time calculations from Stage 3 extended.

NAME _____

CLASS _____

CHAPTER

12

12.01 Ratios

12.02 Equivalent and simplified ratios

12.03 Ratio problems

12.04 Rates

12.05 Best buys

12.06 Rate problems

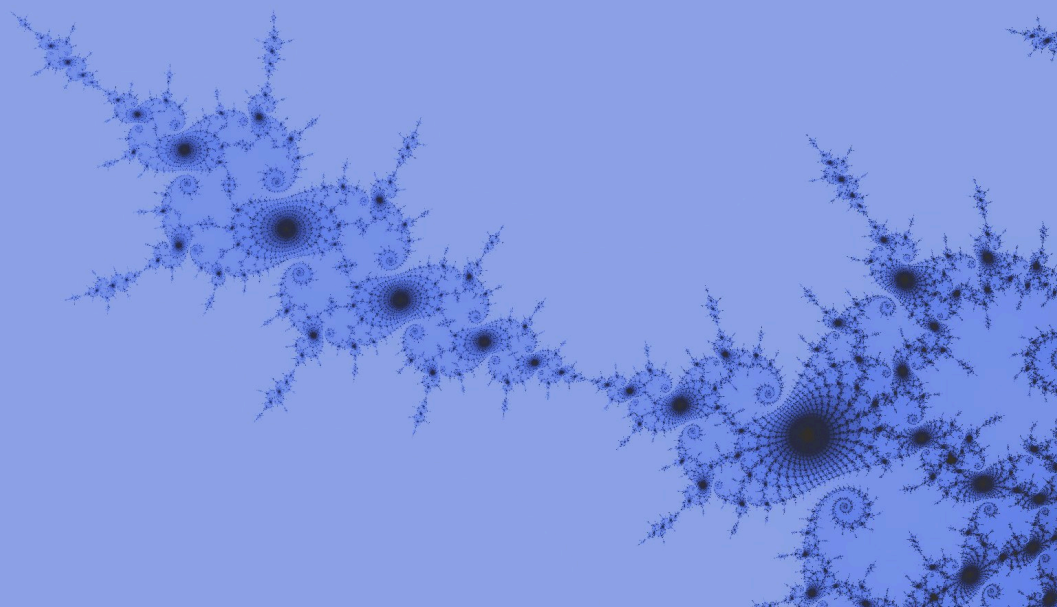
12.07 Travel graphs

12.08 Time

12.09 12-hour and 24-hour time

12.10 Time differences

12.11 Timetables



How to use this booklet

1 Each lesson is **one hour**. Start with the **learning intentions** and **success criteria**, write the **key terms** and use the **notes** box.

WE DO **Grey boxes.** Your teacher works each example (parts a, b, c) on the board. **Copy every step** into the space.

2 Every lesson has three levels: **Easy**, then **Medium**, then **Challenging**. Work down the page: the questions get harder as you go.

YOU DO **Blue boxes.** Now it is your turn. Work on your own. Quick drill answers go on the line; show working in the boxes.

DRILL **Skill drill pages.** After the Easy and the Medium examples there is a full page of short practice questions. Work down the page, write answers on the lines, then mark it and record your score and time.

3 Finished early? Try the **extension**. At the end, **show off your skill** on the tear-off exit ticket and hand it in.

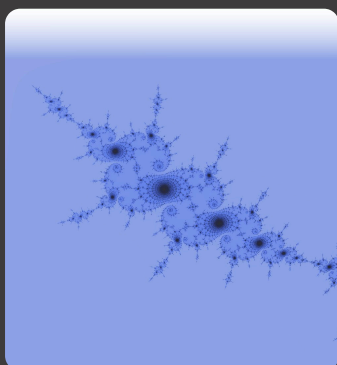
4 Missed a lesson? Each lesson ends with a **summary** with a worked example for every level. Start there.

Contents and progress tracker

Lesson	Topic	Page	Date	Done
12.01	Ratios	3		☐
12.02	Equivalent and simplified ratios	9		☐
12.03	Ratio problems	15		☐
12.04	Rates	21		☐
12.05	Best buys	27		☐
12.06	Rate problems	33		☐
12.07	Travel graphs	39		☐
12.08	Time	47		☐
12.09	12-hour and 24-hour time	53		☐
12.10	Time differences	59		☐
12.11	Timetables	65		☐

In this chapter you will

- write, simplify and use ratios
- solve ratio problems
- write rates and find the best buy
- solve rate problems and read travel graphs
- round, convert and calculate with time
- use 12-hour and 24-hour time, time differences and timetables



About the artwork

The pictures in this booklet are parts of the **Mandelbrot set**, drawn by a computer that repeats one simple rule, $z \rightarrow z^2 + c$, thousands of times for every point. Every booklet shows a different part of the set: this one is a jellyfish of spirals near the neck of the left bulb. Maths can make beautiful things.

Acknowledgement of Country

We acknowledge the Bundjalung people, the Traditional Custodians of the land on which Kingscliff High School stands, and pay our respects to Elders past and present.

LESSON
12.01

Ratios

Date _____

Learning intentions

We are learning to:

- write a ratio to compare quantities

Success criteria

I can:

- ☐ write a ratio in the order the words give
- ☐ use a colon: 3 : 5
- ☐ find a part-to-part and a part-to-whole ratio

Key terms Write the definition as your teacher explains it.

Ratio

Part

Whole

Notes

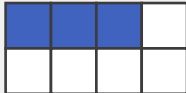
Level 1 · Getting started

Easy · 12 questions

WE DO With your teacher: copy what they write on the board.

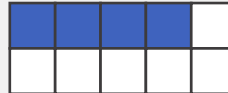
Example 1 Write the ratio of shaded to unshaded parts.

a Ratio of shaded to unshaded?



Your teacher works this out. Copy every step.

b Ratio of shaded to unshaded?



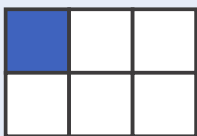
Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** Set A. Write the ratio of shaded to unshaded.

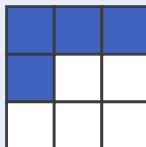
6 QUESTIONS

REMEMBER A ratio compares quantities of the same kind.

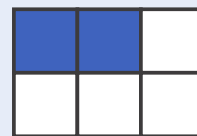
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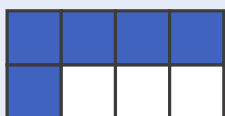
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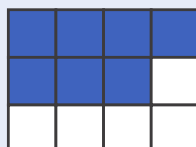
5



2



4



6



SKILL DRILL 1 Easy basics: practise until it is automatic

40 questions

YOU DO

On your own: **Easy** Skill drill 1. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 40

Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–24: write the ratio (do not simplify).

1 2 red, 2 yellow: yellow to red

2 7 blue, 1 green: green to blue

3 4 red, 15 blue: red to blue

4 10 green, 5 blue: blue to green

5 5 blue, 5 yellow: yellow to blue

6 10 green, 8 blue: blue to green

7 15 red, 3 yellow: yellow to red

8 1 red, 15 blue: blue to red

9 7 yellow, 15 green: yellow to green

10 8 yellow, 3 red: red to yellow

11 5 blue, 13 red: blue to all

12 5 yellow, 5 red: yellow to red

13 2 red, 10 yellow: yellow to red

14 7 blue, 3 green: blue to green

15 12 green, 11 yellow: green to all

16 14 green, 15 red: green to red

17 7 red, 9 blue: red to all

18 1 green, 14 yellow: yellow to green

19 10 red, 2 yellow: red to all

20 15 red, 11 yellow: red to all

21 1 yellow, 6 blue: yellow to all

22 15 red, 7 blue: red to all

23 11 blue, 15 red: red to blue

24 11 green, 14 yellow: yellow to green

ROUND 2 Questions 25–40: write the ratio in simplest form.

25 21 girls to 18 boys

26 16 girls to 4 boys

27 8 girls to 12 boys

28 8 girls to 18 boys

29 10 girls to 35 boys

30 25 girls to 5 boys

31 6 girls to 6 boys

32 16 girls to 18 boys

33 40 girls to 5 boys

34 12 girls to 30 boys

35 10 girls to 8 boys

36 3 girls to 27 boys

37 12 girls to 36 boys

38 36 girls to 42 boys

39 8 girls to 8 boys

40 18 girls to 15 boys

YOU DO

On your own: **Easy** Set B. A class has 13 girls and 12 boys. Write the ratio of:

6 QUESTIONS

1 girls to boys

4 boys to the whole class

2 boys to girls

5 teachers (1) to students

3 girls to the whole class

6 students to teachers



Level 2 · Building up

Medium · 6 questions

WE DO

With your teacher: copy what they write on the board.

Example 2

Write each ratio in the order the words give.

a 17 women and 38 men: find the ratio of men to women.

Your teacher works this out. Copy every step.

b Find the ratio of women to the total.

Your teacher works this out. Copy every step.

YOU DO

On your own: **Medium** Set C. Write each ratio.

6 QUESTIONS

REMEMBER Write the parts in the order given.

1 A bag has 7 red and 4 green apples: red to green

3 A team won 9, lost 5, drew 2: wins to losses

5 The letters in RATIO: vowels to consonants

2 Green apples to all apples

4 Losses to games played

6 A 9 m pole and a 4 m pole: short to long

SKILL DRILL 2 Medium basics: keep practising

27 questions

YOU DO

On your own: **Medium** Skill drill 2. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 27 Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–15: change to the same unit, then write the ratio in simplest form.

1 40 c to 1 \$

2 20 mm to 3 cm

3 10 mm to 3 cm

4 10 min to 1 h

5 70 min to 1 h

6 20 g to 4 kg

7 40 cm to 3 m

8 40 min to 2 h

9 90 mm to 4 cm

10 10 min to 5 h

11 20 min to 1 h

12 100 c to 4 \$

13 140 g to 2 kg

14 140 mm to 3 cm

15 90 g to 5 kg

ROUND 2 Questions 16–27: write the ratio of all three in simplest form.

16 25 red, 5 blue, 30 green

17 4 red, 4 blue, 12 green

18 20 red, 10 blue, 25 green

19 8 red, 8 blue, 12 green

20 10 red, 10 blue, 6 green

21 5 red, 5 blue, 10 green

22 6 red, 4 blue, 2 green

23 24 red, 4 blue, 24 green

24 6 red, 18 blue, 12 green

25 3 red, 12 blue, 12 green

26 25 red, 10 blue, 5 green

27 10 red, 8 blue, 8 green

Level 3 · Challenge

Challenging · 4 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Write the ratio, then explain.

- a** A recipe uses 2 cups of flour and 1 cup of sugar. Write the ratio of flour to sugar.

Your teacher works this out. Copy every step.

- b** Is $3 : 5$ the same as $5 : 3$? Explain.

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Think it through.

4 QUESTIONS

REMEMBER Use a colon: $3 : 5$ (read “3 to 5”).

- 1** A drink is 1 part cordial to 4 parts water. What fraction is cordial?

- 3** Draw a 10-square diagram with a shaded to unshaded ratio of $3 : 2$.

- 2** In a ratio of cats to dogs of $3 : 2$, there are 12 cats. How many dogs?

- 4** Can a ratio be written as a fraction? Give an example.

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO **Show off your skill**

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

6 cats and 5 dogs. Write the ratio of cats to dogs.

Level 2

Write the ratio of dogs to all the animals.

Level 3

A juice is 1 part concentrate to 6 parts water. Write the ratio of water to the whole drink.

 Extension · Stretch yourself Worked example

The ratio of boys to girls is 4 : 5. What fraction of the class are boys?

1. Total parts = $4 + 5 = 9$
2. Boys are 4 of the 9 parts

Answer: $\frac{4}{9}$

YOU DO

On your own: read the worked example, then try these.

- E1** The ratio of red to blue to green beads is 2 : 3 : 5. What fraction are blue?

- E2** In a 3 : 7 ratio of juice to water, what percentage is juice?

Lesson summary

Missed the lesson? Start here.

- Remember**
1. A ratio compares quantities of the same kind.
 2. Write the parts in the order given.
 3. Use a colon: 3 : 5 (read “3 to 5”).
 4. Part to whole: add the parts for the whole.

 Level 1

3 shaded, 5 unshaded

1. shaded : unshaded

Answer: 3 : 5

 Level 2

Men to women

1. 38 : 17

Answer: 38 : 17

 Level 3

Women to total

1. 17 : 55

Answer: 17 : 55

 Extension

Boys : girls = 4 : 5

1. 4 of 9 parts

Answer: $\frac{4}{9}$ are boys

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LESSON
12.02

Equivalent and simplified ratios

Date _____

Learning intentions

We are learning to:

- find equivalent ratios and simplify ratios, including with units

Success criteria

I can:

- ☐ multiply or divide every part by the same number
- ☐ divide by the HCF to simplify
- ☐ change to the same units before simplifying

Key terms Write the definition as your teacher explains it.

Equivalent ratio

Simplest form

Notes

Level 1 · Getting started

Easy · 18 questions

WE DO With your teacher: copy what they write on the board.**Example 1** Complete each pair of equivalent ratios.**a** Complete: $2 : 3 = 8 : \underline{\quad}$ **b** Complete: $2 : 3 : 4 = 6 : \underline{\quad} : \underline{\quad}$

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** Set A. Questions 1–6: Complete each pair of equivalent ratios.

12 QUESTIONS

REMEMBER Equivalent: $\times$ or $\div$ every part by the same number.**1** $2 : 3 = 8 : \underline{\quad}$ **2** $1 : 5 = 2 : \underline{\quad}$ **3** $3 : 4 = \underline{\quad} : 20$ **4** $5 : 2 = 15 : \underline{\quad}$ **5** $2 : 3 : 4 = 6 : \underline{\quad} : \underline{\quad}$ **6** $10 : 15 = 2 : \underline{\quad}$

Questions 7–12: simplify.

7 $4 : 6$ **8** $10 : 15$ **9** $8 : 32$ **10** $21 : 14$ **11** $12 : 18 : 30$ **12** $45 : 60$

SKILL DRILL 1 Easy basics: practise until it is automatic

40 questions

YOU DO On your own: **Easy** Skill drill 1. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 40 Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–24: simplify the ratio.

1 42 : 48

2 8 : 36

3 22 : 6

4 12 : 28

5 45 : 36

6 66 : 42

7 14 : 10

8 24 : 22

9 3 : 15

10 8 : 88

11 10 : 18

12 55 : 5

13 36 : 42

14 40 : 96

15 8 : 4

16 14 : 63

17 6 : 66

18 55 : 60

19 16 : 2

20 32 : 20

21 3 : 21

22 18 : 48

23 27 : 45

24 7 : 28

ROUND 2 Questions 25–40: find the missing number.

25 $2 : 2 = \square : 8$

26 $6 : 3 = 36 : \square$

27 $8 : 7 = \square : 56$

28 $6 : 6 = \square : 48$

29 $7 : 2 = 49 : \square$

30 $3 : 9 = 21 : \square$

31 $1 : 7 = \square : 21$

32 $9 : 8 = 36 : \square$

33 $7 : 9 = \square : 36$

34 $8 : 6 = \square : 12$

35 $2 : 1 = 4 : \square$

36 $8 : 6 = 32 : \square$

37 $6 : 8 = 18 : \square$

38 $1 : 9 = \square : 63$

39 $1 : 8 = \square : 64$

40 $3 : 2 = \square : 12$

YOU DO

On your own: **Easy** Set B. Simplify.

6 QUESTIONS

1 12 : 21

4 25 : 45

2 25 : 75

5 18 : 6 : 24

3 6 : 36

6 5 : 25 : 100



Level 2 · Building up

Medium · 6 questions

WE DO

With your teacher: copy what they write on the board.

Example 2

Simplify each ratio.

a Simplify 12 : 21.

Your teacher works this out. Copy every step.

b Simplify 18 : 6 : 24.

Your teacher works this out. Copy every step.

YOU DO

On your own: **Medium** Set C. Change to the same units, then simplify.

6 QUESTIONS

REMEMBER Simplify: ÷ by the HCF of all parts.

1 2 cm : 10 mm

3 3 L : 600 mL

5 40 cents : \$2

2 5 dozen eggs : 6 eggs

4 50 m : 1 km

6 18 months : 3 years

SKILL DRILL 2 Medium basics: keep practising

31 questions

YOU DO

On your own: **Medium** Skill drill 2. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 31 Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–16: simplify. Change decimals or fractions to whole numbers first.

1 $0.3 : 0.7$

2 $0.9 : 0.4$

3 $0.5 : 0.1$

4 $0.5 : 0.6$

5 $0.2 : 0.5$

6 $0.8 : 0.4$

7 $0.6 : 0.4$

8 $0.8 : 0.3$

9 $0.9 : 0.8$

10 $0.5 : 0.8$

11 $0.9 : 0.2$

12 $0.8 : 0.2$

13 $0.6 : 0.3$

14 $0.4 : 0.2$

15 $0.7 : 0.5$

16 $0.6 : 0.1$

ROUND 2 Questions 17–31: simplify the three-part ratio.

17 $2 : 2 : 8$

18 $3 : 6 : 3$

19 $21 : 14 : 28$

20 $3 : 21 : 3$

21 $48 : 8 : 8$

22 $2 : 10 : 12$

23 $20 : 24 : 12$

24 $28 : 7 : 7$

25 $18 : 36 : 12$

26 $14 : 12 : 2$

27 $2 : 12 : 12$

28 $18 : 6 : 9$

29 $48 : 56 : 40$

30 $30 : 30 : 18$

31 $3 : 15 : 12$

Level 3 · Challenge

Challenging · 4 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Change to the same units, then simplify.

a Simplify 2 cm : 10 mm.

Your teacher works this out. Copy every step.

b Simplify 20 minutes : 1 hour.

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Simplify.

4 QUESTIONS

REMEMBER Units: change to the same unit first, then drop the units.

1 20 minutes : 1 hour

3 4 days : 10 hours

2 12 days : 3 weeks

4 1.5 : 2.5 (hint: $\times 2$ first)

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO **Show off your skill**

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Complete: $3 : 7 = 9 : \underline{\quad}$

Level 2

Simplify 16 : 40.

Level 3

Simplify 30 cm : 2 m.

Extension · Stretch yourself

Worked example

Simplify $\frac{1}{2} : \frac{1}{3}$.

1. Multiply both by 6 (the LCD)
2. $3 : 2$

Answer: $3 : 2$

YOU DO

On your own: read the worked example, then try these.

E1 Simplify $\frac{3}{4} : \frac{5}{8}$.

E2 Simplify $0.25 \text{ kg} : 750 \text{ g} : 1.5 \text{ kg}$.

Lesson summary

Missed the lesson? Start here.

Remember

1. Equivalent: $\times$ or $\div$ every part by the same number.
2. Simplify: $\div$ by the HCF of all parts.
3. Units: change to the same unit first, then drop the units.
4. Decimals or fractions: multiply to make whole numbers.

Level 1

$2 : 3 = 8 : ?$

1. $\times 4$

Answer: $8 : 12$

Level 2

$12 : 21$

1. $\div 3$

Answer: $4 : 7$

Level 3

$2 \text{ cm} : 10 \text{ mm}$

1. $20 \text{ mm} : 10 \text{ mm}$

Answer: $2 : 1$

Extension

$\frac{1}{2} : \frac{1}{3}$

1. $\times 6$

Answer: $3 : 2$

This space is left blank on purpose. It is the back of your exit ticket.

LESSON 12.03

Ratio problems

Date _____

Learning intentions

We are learning to:

- solve problems using ratios

Success criteria

I can:

- ☐ use the unitary method: find one part first
- ☐ find an unknown quantity from a ratio
- ☐ divide a quantity in a given ratio

Key terms Write the definition as your teacher explains it.

Unitary method

Share

Notes

Level 1 · Getting started

Easy · 18 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Use the ratio to find the unknown amount.

- a** Girls : boys = 4 : 3. There are 75 boys. How many girls?
- b** Tyrone : Erin invested in the ratio 5 : 7. Erin invested \$63 000. How much did Tyrone invest?

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** Set A. Questions 1–6: Find the unknown.

12 QUESTIONS

REMEMBER Find the value of one part.

① $3:4 = 12:?$

2 $2 : 5 = ? : 30$

3 $7 : 2 = 35 : ?$

④ $1:6 = 8:?$

5 $5:3 = ? : 12$

6 $4:9 = 20:?$

Questions 7–12: share \$60 in each ratio.

⑦ 1 : 1

⑧ 1:2

⑨ 1:3

10 2:3

11 1:5

12 7:5

SKILL DRILL 1 Easy basics: practise until it is automatic

36 questions

YOU DO

On your own: **Easy** Skill drill 1. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 36

Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–24: share the amount in the ratio given.

1 \$35 in 4 : 3

2 \$55 in 4 : 1

3 \$90 in 3 : 2

4 \$12 in 1 : 3

5 \$90 in 1 : 5

6 \$20 in 4 : 1

7 \$44 in 3 : 1

8 \$30 in 1 : 4

9 \$56 in 3 : 4

10 \$24 in 2 : 1

11 \$120 in 4 : 2

12 \$60 in 1 : 3

13 \$54 in 2 : 1

14 \$12 in 1 : 2

15 \$14 in 5 : 2

16 \$32 in 1 : 3

17 \$70 in 4 : 3

18 \$90 in 5 : 5

19 \$45 in 2 : 1

20 \$27 in 2 : 1

21 \$36 in 3 : 3

22 \$126 in 3 : 4

23 \$6 in 1 : 2

24 \$21 in 2 : 1

ROUND 2 Questions 25–36: the ratio and one amount are given. Find the other.

25 boys : girls = 2 : 4. There are 14 boys. How many girls?

26 boys : girls = 4 : 2. There are 16 boys. How many girls?

27 boys : girls = 5 : 3. There are 45 boys. How many girls?

28 boys : girls = 1 : 2. There are 8 boys. How many girls?

29 boys : girls = 6 : 4. There are 48 boys. How many girls?

30 boys : girls = 1 : 3. There are 6 boys. How many girls?

31 boys : girls = 1 : 4. There are 8 boys. How many girls?

32 boys : girls = 2 : 1. There are 12 boys. How many girls?

33 boys : girls = 3 : 5. There are 27 boys. How many girls?

34 boys : girls = 5 : 3. There are 35 boys. How many girls?

35 boys : girls = 4 : 3. There are 16 boys. How many girls?

36 boys : girls = 3 : 1. There are 21 boys. How many girls?

YOU DO

On your own: **Easy** Set B. Share each amount in the ratio given.

6 QUESTIONS

1 \$60 in 1 : 2

4 72 lollies in 5 : 3

2 \$100 in 2 : 3

5 \$1000 in 1 : 4

3 \$45 in 4 : 5

6 48 kg in 1 : 2 : 3



Level 2 · Building up

Medium · 6 questions

WE DO

With your teacher: copy what they write on the board.

Example 2

Share the amount in the ratio.

a Share \$60 in the ratio 1 : 2.

Your teacher works this out. Copy every step.

b Share \$450 in the ratio 4 : 5.

Your teacher works this out. Copy every step.

YOU DO

On your own: **Medium** Set C. Solve.

6 QUESTIONS

REMEMBER Multiply to find the other quantities.

1 Girls : boys = 4 : 3. There are 75 boys. How many girls?

3 Paint is mixed blue : white = 2 : 5. How much white for 6 L of blue?

5 Map scale 1 : 50 000. 3 cm on the map is how many km?

2 Tyrone : Erin = 5 : 7. Erin invested \$63 000. Find Tyrone's share.

4 Share \$450 in the ratio 4 : 5.

6 Cement : sand : gravel = 1 : 2 : 3. How much sand in 36 kg?

SKILL DRILL 2 Medium basics: keep practising

25 questions

YOU DO

On your own: **Medium** Skill drill 2. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 25

Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–15: share the quantity in the ratio given.

1 60 kg in 4 : 3 : 3

2 12 L in 3 : 2 : 1

3 63 L in 2 : 4 : 3

4 15 kg in 2 : 2 : 1

5 80 L in 4 : 5 : 1

6 72 kg in 3 : 2 : 1

7 49 m in 1 : 4 : 2

8 64 L in 4 : 3 : 1

9 77 m in 1 : 4 : 2

10 63 kg in 4 : 2 : 1

11 63 kg in 4 : 2 : 3

12 50 m in 2 : 2 : 1

13 72 m in 1 : 5 : 3

14 120 m in 3 : 5 : 2

15 35 kg in 2 : 3 : 2

ROUND 2 Questions 16–25: the difference between the two shares is given. Find each share.

16 ratio 4 : 7; the larger share is \$33 more

17 ratio 2 : 4; the larger share is \$6 more

18 ratio 4 : 5; the larger share is \$12 more

19 ratio 4 : 5; the larger share is \$5 more

20 ratio 1 : 2; the larger share is \$6 more

21 ratio 2 : 6; the larger share is \$40 more

22 ratio 4 : 5; the larger share is \$13 more

23 ratio 1 : 2; the larger share is \$7 more

24 ratio 3 : 6; the larger share is \$39 more

25 ratio 4 : 8; the larger share is \$48 more

Level 3 · Challenge

Challenging · 4 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Solve the ratio problem.

- a** A map has scale 1 : 50 000. 3 cm on the map is how far in real life?

Your teacher works this out. Copy every step.

- b** Cement, sand and gravel are mixed 1 : 2 : 3. How much sand is in 36 kg of mix?

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Harder problems.

4 QUESTIONS

REMEMBER To share: total parts, then amount ÷ total parts.

- 1** The ratio of adults to children at a show is 3 : 8. There were 440 people. How many were children?

- 3** A recipe for 4 people uses 300 g of pasta. How much for 10 people?

- 2** Sam and Jo share money 3 : 5. Jo gets \$24 more than Sam. How much does each get?

- 4** A photo 10 cm by 15 cm is enlarged in the ratio 1 : 3. Find the new size.

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO **Show off your skill**

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Complete: 2 : 7 = 10 : ____

Level 2

Share \$80 in the ratio 3 : 5.

Level 3

Juice : water = 1 : 4. How much water for 250 mL of juice?

 Extension · Stretch yourself Worked example

The ratio of red to blue marbles is 3 : 5. After 6 red marbles are added, the ratio is 1 : 1. How many blue marbles are there?

1. Red 3k, blue 5k
2. $3k + 6 = 5k$, so $2k = 6$, $k = 3$

Answer: 15 blue

YOU DO

On your own: read the worked example, then try these.

- E1 The ratio of boys to girls is 2 : 3. After 4 boys join, it is 1 : 1. How many girls?

- E2 Share \$84 among A, B and C so that A gets twice B and C gets three times B.

Lesson summary

Missed the lesson? Start here.

- Remember**
1. Find the value of one part.
 2. Multiply to find the other quantities.
 3. To share: total parts, then amount $\div$ total parts.
 4. Check the parts add back to the total.

 Level 1

Girls : boys = 4 : 3, boys 75

1. 1 part = 25

Answer: 100 girls

 Level 2

Tyrone : Erin = 5 : 7

1. 1 part = $63\,000 \div 7 = 9000$

Answer: \$45 000

 Level 3

\$450 in 4 : 5

1. 9 parts, 1 part = \$50

Answer: \$200 and \$250

 Extension

Marbles 3 : 5 $\rightarrow$ 1 : 1

1. $2k = 6$

Answer: 15 blue

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
12.04

Rates

Date _____

Learning intentions

We are learning to:

- write and simplify rates

Success criteria

I can:

- ☐ know a rate compares quantities of different kinds
- ☐ write a rate with both units, e.g. km/h
- ☐ simplify a rate to a rate per 1 unit

Key terms Write the definition as your teacher explains it.

Rate

Unit rate

Notes

Level 1 · Getting started

Easy · 18 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Write each as a rate in simplest form.

- a** Write as a rate: 150 000 voters for 50 politicians.

Your teacher works this out. Copy every step.

- b** Write as a rate: 1000 sheep over 5 hectares.

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** Set A. Questions 1–6: Write each as a rate in simplest form.

12 QUESTIONS

REMEMBER A **RATE** compares different kinds of quantities.

- 1** 150 000 voters for 50 politicians

- 2** 1000 sheep over 5 hectares

- 3** 10 tickets for \$8

- 4** 810 stitches for 3 balls

- 5** \$120 for 8 hours

- 6** 240 km in 3 hours

Questions 7–12: write as a rate per 1 unit.

- 7** \$20 for 4 kg

- 8** 300 km in 5 h

- 9** 45 L in 9 min

- 10** \$36 for 3 h

- 11** 120 words in 2 min

- 12** \$8 for 10 pens

SKILL DRILL 1 Easy basics: practise until it is automatic

36 questions

YOU DO

On your own: **Easy** Skill drill 1. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 36

Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–24: write as a rate for one unit.

1 \$165 in 3 kg

2 72 words in 4 min

3 15 words in 5 min

4 156 beats in 6 min

5 35 L in 5 min

6 306 words in 6 min

7 \$108 in 9 kg

8 18 km in 3 h

9 133 m in 7 s

10 432 km in 8 h

11 114 L in 3 min

12 240 beats in 4 min

13 320 km in 8 h

14 120 m in 3 s

15 140 beats in 5 min

16 72 beats in 4 min

17 32 beats in 4 min

18 228 km in 4 h

19 210 km in 6 h

20 112 words in 4 min

21 \$91 in 7 kg

22 48 km in 8 h

23 288 words in 8 min

24 \$168 in 3 kg

ROUND 2 Questions 25–36: what unit would you use for each rate?

25 typing speed

26 running speed

27 wage

28 growth of a plant

29 petrol price

30 phone data

31 water from a tap

32 price of apples

33 rainfall

34 fuel use

35 heart rate

36 speed of a car

YOU DO

On your own: **Easy** Set B. Write each as a rate per 1 unit.

6 QUESTIONS

- | | | | |
|----------------------|----------------------|----------------------------------|----------------------|
| 1 \$15 for 3 kg | <input type="text"/> | 4 72 beats in 1 min (per second) | <input type="text"/> |
| 2 60 L in 4 minutes | <input type="text"/> | 5 \$4.50 for 5 L | <input type="text"/> |
| 3 450 words in 5 min | <input type="text"/> | 6 56 km on 4 L | <input type="text"/> |



Level 2 · Building up

Medium · 6 questions

WE DO

With your teacher: copy what they write on the board.

Example 2

Write each as a rate in simplest form.

- a** Write as a rate: 10 tickets for \$8.

Your teacher works this out. Copy every step.

- b** Write as a rate: 810 stitches to make 3 balls.

Your teacher works this out. Copy every step.

YOU DO

On your own: **Medium** Set C. Find each rate. Include units.

6 QUESTIONS

REMEMBER Write both units: km/h, \$/kg, L/min.

- | | | |
|--|--|--|
| 1 A car travels 240 km in 3 hours. Speed? | 3 A worker earns \$187 for 8.5 hours. Hourly rate? | 5 A printer prints 150 pages in 6 min. Pages per minute? |
| <div style="border: 1px solid black; height: 180px; width: 100%;"></div> | <div style="border: 1px solid black; height: 180px; width: 100%;"></div> | <div style="border: 1px solid black; height: 180px; width: 100%;"></div> |
| 2 A tap fills 45 L in 9 minutes. Rate? | 4 A plane flies 3600 km in 4.5 hours. Speed? | 6 A runner runs 400 m in 80 s. Speed in m/s? |
| <div style="border: 1px solid black; height: 180px; width: 100%;"></div> | <div style="border: 1px solid black; height: 180px; width: 100%;"></div> | <div style="border: 1px solid black; height: 180px; width: 100%;"></div> |

SKILL DRILL 2 Medium basics: keep practising

24 questions

YOU DO

On your own: **Medium** Skill drill 2. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 24

Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–12: convert the rate.

1 6 km/h to km/min

2 120 km/h to km/min

3 120 km/min to km/h

4 30 km/min to km/h

5 \$36/kg to c/g

6 18 km/h to km/min

7 \$60/kg to c/g

8 90 km/min to km/h

9 36 km/h to km/min

10 60 km/min to km/h

11 \$6/kg to c/g

12 90 km/h to km/min

ROUND 2 Questions 13–24: which rate is faster (or cheaper)? Change both to the same unit.

13 A: 47 km/h, B: 394 km in 4 h

14 A: 44 km/h, B: 163 km in 5 h

15 A: 36 km/h, B: 253 km in 4 h

16 A: 53 km/h, B: 145 km in 4 h

17 A: 33 km/h, B: 149 km in 5 h

18 A: 72 km/h, B: 316 km in 2 h

19 A: 59 km/h, B: 187 km in 5 h

20 A: 69 km/h, B: 255 km in 4 h

21 A: 71 km/h, B: 356 km in 3 h

22 A: 62 km/h, B: 228 km in 2 h

23 A: 73 km/h, B: 295 km in 2 h

24 A: 58 km/h, B: 342 km in 4 h



Level 3 · Challenge

Challenging · 4 questions · then try the extension

WE DO With your teacher: copy what they write on the board.**Example 3** Find the speed, then convert the rate.

- a** A car travels 240 km in 3 hours. Find the speed in km/h.

Your teacher works this out. Copy every step.

- b** Convert 90 km/h to metres per minute.

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Converting rates.

4 QUESTIONS

REMEMBER Unit rate: divide to get “per 1”.

- 1** Convert 90 km/h to metres per minute.

- 3** Convert \$18 per hour to cents per minute.

- 2** Convert 5 m/s to km/h.

- 4** Which is faster: 20 m/s or 70 km/h?

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO Show off your skillChoose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Write \$36 for 4 kg as a rate per kg.

Level 2

A car travels 150 km in 2 h. Find its speed.

Level 3

Convert 60 km/h to metres per minute.

 Extension · Stretch yourself Worked example

Convert 72 km/h to m/s.

1. $72 \text{ km} = 72\,000 \text{ m}$; $1 \text{ h} = 3600 \text{ s}$
2. $72\,000 \div 3600$

Answer: 20 m/s

YOU DO

On your own: read the worked example, then try these.

- E1 Convert 15 m/s to km/h.

- E2 A snail moves at 3 mm/s. How many metres per hour is that?

Lesson summary

Missed the lesson? Start here.

- Remember**
1. A **rate** compares different kinds of quantities.
 2. Write both units: km/h, \$/kg, L/min.
 3. Unit rate: divide to get “per 1”.
 4. Convert units on the top and bottom separately.

 Level 1

150 000 voters, 50 politicians

1. $\div 50$

Answer: 3000 voters/politician

 Level 2

10 tickets for \$8

1. $8 \div 10$

Answer: \$0.80/ticket

 Level 3

240 km in 3 h

1. $240 \div 3$

Answer: 80 km/h

 Extension

72 km/h to m/s

1. $72\,000 \div 3600$

Answer: 20 m/s

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
12.05

Best buys

Date _____

Learning intentions

We are learning to:

- compare prices to find the best buy

Success criteria

I can:

- ☐ find the unit price (price per 1 unit or per 100 g)
- ☐ change to the same units before comparing
- ☐ decide the best buy and give a reason

Key terms Write the definition as your teacher explains it.**Unit price****Best buy**

Notes

Level 1 · Getting started

Easy · 18 questions

WE DO With your teacher: copy what they write on the board.**Example 1** Find the price per 100 g, then choose the better buy.

- a** Dog biscuits: 400 g of Woof for \$5.20. Find the price per 100 g.

Your teacher works this out. Copy every step.

- b** 1 kg of Chow for \$11.10. Find the price per 100 g. Which is the better buy?

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** Set A. Questions 1–6: Find the unit price (price per 1 item or 1 unit). 12 QUESTIONS**REMEMBER** Find the price for the same amount (per 1 or per 100 g).

- 1 4 apples for \$2.40

- 2 6 eggs for \$3.90

- 3 2 kg for \$7

- 4 5 L for \$8.50

- 5 12 pens for \$9

- 6 3 m for \$4.20

Questions 7–12: find the unit price.

- 7 4 for \$6

- 8 2 kg for \$9

- 9 5 L for \$12

- 10 10 for \$4.50

- 11 3 m for \$7.50

- 12 6 for \$9.60

SKILL DRILL 1 Easy basics: practise until it is automatic

36 questions

YOU DO

On your own: **Easy** Skill drill 1. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 36

Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–24: find the unit price (the cost of one).

1 7 for \$26.18

2 4 for \$35.00

3 4 for \$8.92

4 10 for \$70.60

5 3 for \$21.12

6 5 for \$5.55

7 3 for \$19.02

8 3 for \$11.43

9 6 for \$51.78

10 11 for \$68.31

11 8 for \$68.48

12 6 for \$22.68

13 8 for \$62.16

14 5 for \$25.70

15 6 for \$7.86

16 5 for \$16.70

17 7 for \$20.65

18 6 for \$38.82

19 6 for \$48.24

20 5 for \$44.05

21 8 for \$7.36

22 10 for \$52.80

23 8 for \$37.04

24 9 for \$29.34

ROUND 2 Questions 25–36: which is the better buy? Find the price of 1 of each.

25 4 for \$15.56 or 3 for \$11.94

26 3 for \$11.70 or 5 for \$5.10

27 6 for \$23.40 or 2 for \$4.68

28 2 for \$5.70 or 3 for \$9.48

29 3 for \$10.02 or 5 for \$12.75

30 2 for \$3.44 or 6 for \$6.84

31 4 for \$15.08 or 3 for \$9.06

32 4 for \$4.08 or 6 for \$8.52

33 2 for \$2.60 or 4 for \$13.72

34 6 for \$8.94 or 4 for \$10.80

35 6 for \$13.92 or 4 for \$15.56

36 4 for \$8.04 or 5 for \$12.45

YOU DO On your own: **Easy** Set B. Find the price per 100 g.

6 QUESTIONS

1 400 g for \$5.20

4 500 g for \$4.25

2 1 kg for \$11.10

5 750 g for \$6

3 250 g for \$3

6 2 kg for \$19

Level 2 · Building up

Medium · 6 questions

WE DO With your teacher: copy what they write on the board.

Example 2 Which is the better buy?

a Milk: 1 L of Dairy Farm for \$2.85 or 1250 mL of Milky for \$3.90. Which is better value?

Your teacher works this out. Copy every step.

b Juice: 2 L for \$4.60 or 3 L for \$6.60. Which is better value?

Your teacher works this out. Copy every step.

YOU DO On your own: **Medium** Set C. Which is the better buy? Show working.

6 QUESTIONS

REMEMBER Change units first (1 kg = 1000 g, 1 L = 1000 mL).

1 400 g of Woof for \$5.20 or 1 kg of Chow for \$11.10

3 2 L for \$4.60 or 3 L for \$6.60

5 6 rolls for \$4.80 or 10 rolls for \$7.50

2 1 L for \$2.85 or 1250 mL for \$3.90

4 500 g for \$3.75 or 800 g for \$5.60

6 375 mL for \$1.50 or 1.25 L for \$4.50

SKILL DRILL 2 Medium basics: keep practising

27 questions

YOU DO

On your own: **Medium** Skill drill 2. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 27 Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–12: which is the better buy? Find the price per 100 g (or per 100 mL).

1 375 g for \$5.44 or 750 g for \$5.72

2 150 g for \$5.96 or 600 g for \$9.28

3 375 g for \$2.90 or 750 g for \$6.07

4 250 g for \$1.67 or 500 g for \$9.41

5 150 g for \$1.90 or 600 g for \$7.16

6 250 g for \$3.00 or 500 g for \$7.56

7 150 g for \$5.67 or 600 g for \$11.40

8 300 g for \$3.66 or 1000 g for \$7.53

9 375 g for \$5.27 or 750 g for \$3.52

10 400 g for \$3.37 or 1000 g for \$3.68

11 150 g for \$2.14 or 600 g for \$8.65

12 150 g for \$1.93 or 600 g for \$7.43

ROUND 2 Questions 13–27: find the cost.

13 0.5 kg at \$5.36/kg

14 0.5 kg at \$13.26/kg

15 1.5 kg at \$10.24/kg

16 2.5 kg at \$2.72/kg

17 6 kg at \$7.73/kg

18 5 kg at \$8.97/kg

19 2.5 kg at \$5.72/kg

20 5 kg at \$12.17/kg

21 3 kg at \$3.62/kg

22 1.5 kg at \$2.87/kg

23 1.5 kg at \$6.45/kg

24 4 kg at \$2.26/kg

25 6 kg at \$14.20/kg

26 2.5 kg at \$9.80/kg

27 10 kg at \$4.31/kg

Level 3 · Challenge

Challenging · 4 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Which is the better buy? Explain.

a Why might someone not choose the best buy?

Your teacher works this out. Copy every step.

b Pens: 3 for \$4.20 or 5 for \$6.50. Which is cheaper per pen?

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Problems.

4 QUESTIONS

REMEMBER The lower unit price is the better buy.

1 Rice: 1 kg for \$2.40, 2 kg for \$4.50, 5 kg for \$10.50. Which is the best buy?

3 Is buying in bulk always the best choice? Explain.

2 A 600 g jar costs \$4.20. A 1 kg jar costs \$6.50. How much do you save per 100 g with the bigger jar?

4 Find a best buy at a supermarket (online). Show the unit prices.

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO **Show off your skill**

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Find the price per kg: 3 kg for \$12.60.

Level 2

Which is better: 1 L for \$3 or 2 L for \$5.40?

Level 3

Which is better: 250 g for \$2.50 or 1 kg for \$9?

Extension · Stretch yourself**Worked example**

A shop has “3 for \$10” or \$3.60 each. How much do you save buying 6?

1. 6 at \$3.60 = \$21.60
2. 2 lots of 3 for \$10 = \$20

Answer: \$1.60

YOU DO

On your own: read the worked example, then try these.

- E1** Shampoo: 400 mL for \$6 or 1 L for \$13.50. Which is better? By how much per 100 mL?

- E2** A “buy 2 get 1 free” deal on \$4.50 items: find the price per item.

Lesson summary

Missed the lesson? Start here.

- Remember**
1. Find the price for the same amount (per 1 or per 100 g).
 2. Change units first (1 kg = 1000 g, 1 L = 1000 mL).
 3. The lower unit price is the better buy.
 4. Consider waste, storage and need.

Level 1

Woof

1. $5.20 \div 4$

Answer: \$1.30/100 g

Level 2

Chow

1. $11.10 \div 10$

Answer: \$1.11/100 g: better

Level 3

Milk

1. \$2.85/L vs \$3.12/L

Answer: Dairy Farm

Extension

3 for \$10 vs \$3.60 each (6)

1. \$20 vs \$21.60

Answer: save \$1.60

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
12.06

Rate problems

Date _____

Learning intentions

We are learning to:

- solve problems using rates

Success criteria

I can:

- ☐ multiply a rate by a quantity to find a total
- ☐ divide by a rate to find how many
- ☐ round sensibly

Key terms Write the definition as your teacher explains it.**Rate****Unitary method**

Notes

Level 1 • Getting started

Easy • 18 questions

WE DO With your teacher: copy what they write on the board.**Example 1** Use the rate to find the total.

- a** Marie earns \$22.45 per hour. How much for 38 hours?

Your teacher works this out. Copy every step.

- b** Petrol costs \$1.79/L. How many litres can you buy for \$65 (1 d.p.)?

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** Set A. Questions 1–6: Use the rate to find the total.

12 QUESTIONS

REMEMBER Total = rate $\times$ amount.

- 1 \$22.45/h for 38 h

- 2 80 km/h for 2.5 h

- 3 \$3.50/kg for 4 kg

- 4 15 L/min for 8 min

- 5 60 words/min for 12 min

- 6 \$1.79/L for 40 L

Questions 7–12: use the rate.

- 7 \$15/h for 4 h

- 8 60 km/h for 3 h

- 9 \$2/kg for 7 kg

- 10 10 L/min for 6 min

- 11 \$12 at \$3/kg (kg?)

- 12 200 km at 50 km/h (h?)

SKILL DRILL 1 Easy basics: practise until it is automatic

40 questions

YOU DO

On your own: **Easy** Skill drill 1. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 40

Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–24: use the rate to find the total.

1 6 kg at \$25/kg

2 3 kg at \$12/kg

3 11 kg at \$19/kg

4 4 h at 70 km/h

5 5 kg at \$15/kg

6 7 kg at \$16/kg

7 6 kg at \$23/kg

8 9 h at 105 km/h

9 4 h at \$20/h

10 2 h at \$11/h

11 3 h at \$4/h

12 9 h at \$16/h

13 9 h at \$4/h

14 8 h at \$11/h

15 12 h at 70 km/h

16 2 h at \$16/h

17 6 h at \$7/h

18 9 kg at \$15/kg

19 10 kg at \$17/kg

20 12 kg at \$4/kg

21 8 kg at \$7/kg

22 5 h at 15 km/h

23 11 h at \$9/h

24 11 h at 20 km/h

ROUND 2 Questions 25–40: find the time taken: $\text{time} = \text{distance} \div \text{speed}$.

25 400 km at 80 km/h

26 200 km at 50 km/h

27 100 km at 50 km/h

28 250 km at 50 km/h

29 160 km at 80 km/h

30 120 km at 40 km/h

31 150 km at 50 km/h

32 80 km at 80 km/h

33 60 km at 60 km/h

34 300 km at 100 km/h

35 300 km at 50 km/h

36 320 km at 80 km/h

37 200 km at 100 km/h

38 180 km at 60 km/h

39 500 km at 100 km/h

40 240 km at 40 km/h

YOU DO On your own: **Easy** Set B. Find how many or how long.

6 QUESTIONS

- | | | | |
|-----------------------------|----------------------|-----------------------------|----------------------|
| 1 \$65 at \$1.79/L (1 d.p.) | <input type="text"/> | 4 360 pages at 24 pages/min | <input type="text"/> |
| 2 150 kg at 20 kg/ha | <input type="text"/> | 5 \$90 at \$7.50/h | <input type="text"/> |
| 3 210 km at 70 km/h | <input type="text"/> | 6 1000 L at 40 L/min | <input type="text"/> |

Level 2 · Building up

Medium · 6 questions

WE DO With your teacher: copy what they write on the board.

Example 2 Use the rate to find how many.

- a** Fertiliser is used at 20 kg/ha. How many hectares does 150 kg cover?
- b** A car travels at 80 km/h for 2.5 h. How far does it go?

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Medium** Set C. Solve.

6 QUESTIONS

REMEMBER How many = total $\div$ rate.

- | | | |
|---|--|---|
| 1 A tap drips 12 mL per minute. How many litres per day? | 3 A cyclist rides at 24 km/h. How far in 45 minutes? | 5 A heart beats 72 times a minute. How many beats in a day? |
| <input type="text"/> | <input type="text"/> | <input type="text"/> |
| 2 A plumber charges \$85 call-out plus \$95/h. Find the cost for 3.5 h. | 4 Grass seed is spread at 35 g/m ² . How much for a 12 m by 8 m lawn? | 6 A car uses 7.5 L/100 km. How much fuel for 360 km? |
| <input type="text"/> | <input type="text"/> | <input type="text"/> |

SKILL DRILL 2 Medium basics: keep practising

24 questions

YOU DO

On your own: **Medium** Skill drill 2. Work down the page, one round at a time. Write each answer on its line (show working in the box).

Start time _____ Finish time _____ Score _____ / 24 Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–15: find the distance: distance = speed × time.1 60 km/h for 30 min
_____2 90 km/h for 15 min
_____3 60 km/h for 1 h 30 min
_____4 100 km/h for 30 min
_____5 90 km/h for 45 min
_____6 80 km/h for 30 min
_____7 60 km/h for 45 min
_____8 120 km/h for 2 h 30 min
_____9 90 km/h for 1 h 30 min
_____10 80 km/h for 1 h 30 min
_____11 40 km/h for 1 h 30 min
_____12 90 km/h for 2 h 30 min
_____13 120 km/h for 45 min
_____14 100 km/h for 45 min
_____15 120 km/h for 1 h 30 min
_____**ROUND 2** Questions 16–24: two steps. Show your working.16 Mia earns \$29/h. She works 6 h a day for 5 days. How much does she earn?
17 Mia earns \$15/h. She works 8 h a day for 5 days. How much does she earn?
18 Mia earns \$26/h. She works 3 h a day for 5 days. How much does she earn?
19 Mia earns \$30/h. She works 8 h a day for 5 days. How much does she earn?
20 Mia earns \$29/h. She works 7 h a day for 3 days. How much does she earn?
21 Mia earns \$18/h. She works 8 h a day for 2 days. How much does she earn?
22 Mia earns \$15/h. She works 3 h a day for 3 days. How much does she earn?
23 Mia earns \$29/h. She works 5 h a day for 4 days. How much does she earn?
24 Mia earns \$27/h. She works 7 h a day for 3 days. How much does she earn?

Level 3 · Challenge

Challenging · 4 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Use the rate to find the time or distance.

- a** How long does it take to travel 210 km at 70 km/h?

- b** A tap drips 12 mL per minute. How many litres in a day?

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Harder problems.

4 QUESTIONS

REMEMBER Time = distance ÷ speed.

- 1** Two taps fill a tank: one at 20 L/min, the other at 30 L/min. How long to fill 1500 L together?

- 3** Ann types 45 words/min and Ben 60 words/min. How much longer does Ann take to type 1800 words?

- 2** A car uses 8 L/100 km. Petrol is \$1.85/L. Find the fuel cost of a 450 km trip.

- 4** Which is faster: 100 m in 12.5 s or 1 km in 2 min 30 s?

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO **Show off your skill**

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Find the pay for 6 h at \$24.50/h.

Level 2

How long to travel 180 km at 60 km/h?

Level 3

How many litres of petrol for \$50 at \$1.60/L?

 Extension · Stretch yourself Worked example

A train leaves at 9 a.m. travelling at 90 km/h. A car leaves the same place at 10 a.m. at 120 km/h on the same road. When does the car catch up?

1. At 10 a.m. the train is 90 km ahead
2. The car gains 30 km every hour
3. $90 \div 30 = 3$ hours after 10 a.m.

Answer: 1 p.m.

YOU DO

On your own: read the worked example, then try these.

- E1 How far from the start does the car catch up?

- E2 Two people walk towards each other from 12 km apart at 4 km/h and 2 km/h. When do they meet?

Lesson summary

Missed the lesson? Start here.

Remember

1. Total = rate $\times$ amount.
2. How many = total $\div$ rate.
3. Time = distance $\div$ speed.
4. Check units match (h and min, L and mL).

 Level 1

$$\$22.45/\text{h} \times 38 \text{ h}$$

1. 22.45×38

Answer: \$853.10

 Level 2

$$\$65 \div \$1.79/\text{L}$$

1. 36.31...

Answer: 36.3 L

 Level 3

$$150 \text{ kg} \div 20 \text{ kg/ha}$$

1. 7.5

Answer: 7.5 ha

 Extension

Car catches train

1. $90 \div 30 = 3 \text{ h}$

Answer: 1 p.m.

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
12.07

Travel graphs

Date _____

Learning intentions

We are learning to:

- read and interpret travel (distance–time) graphs

Success criteria

I can:

- ☐ read distance and time from a travel graph
- ☐ identify stops (flat sections) and the return journey
- ☐ find speed = distance ÷ time for a section

Key terms Write the definition as your teacher explains it.

Travel graph

Speed

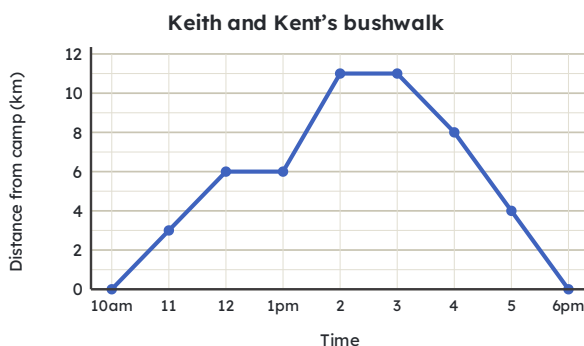
Notes

Level 1 · Getting started

Easy · 14 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Use the travel graph.



a How far did Keith and Kent walk altogether?

Your teacher works this out. Copy every step.

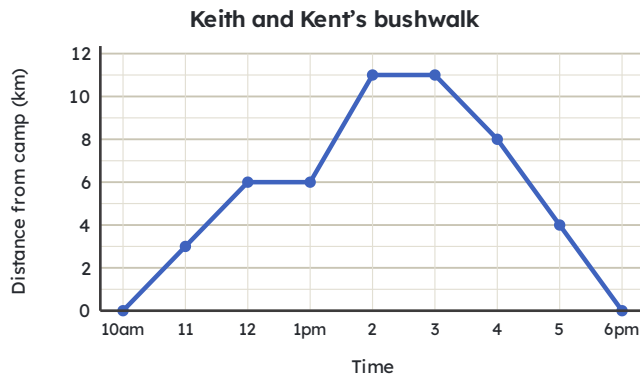
b How many stops did they make?

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** Set A. Use the travel graph.

4 QUESTIONS

REMEMBER Time is across, distance is up.



1 Total distance walked?

2 Number of stops?

3 Time they started back?

4 Fastest section?

YOU DO On your own: **Easy** Set A2. Quick drill: find the speed.

6 QUESTIONS

1 60 km in 2 h

2 12 km in 3 h

3 150 km in 1.5 h

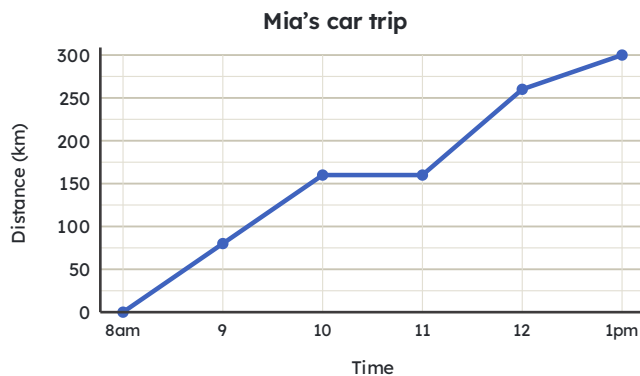
4 5 km in 30 min

5 0 km in 1 h

6 240 km in 4 h

YOU DO On your own: **Easy** Set B. Use the travel graph.

4 QUESTIONS



1 How far did Mia drive?

2 How long did she stop?

3 Speed from 8 to 10 a.m.?

4 Speed from 12 to 1 p.m.?

SKILL DRILL 1 Easy basics: practise until it is automatic

27 questions

YOU DO On your own: **Easy** Skill drill 1. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 27 Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–21: find the speed: $\text{speed} = \text{distance} \div \text{time}$.

1 75 km in 5 h

2 400 km in 4 h

3 20 km in 2 h

4 90 km in 2 h

5 480 km in 6 h

6 200 km in 5 h

7 180 km in 4 h

8 120 km in 2 h

9 50 km in 5 h

10 200 km in 2 h

11 240 km in 6 h

12 360 km in 4 h

13 540 km in 6 h

14 300 km in 3 h

15 120 km in 3 h

16 100 km in 5 h

17 60 km in 3 h

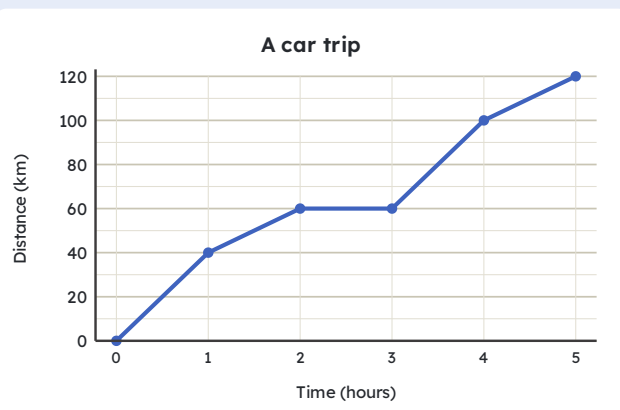
18 450 km in 5 h

19 180 km in 3 h

20 300 km in 5 h

21 40 km in 4 h

ROUND 2 Questions 22–27: use the travel graph.



22 How far was travelled in the first hour?

23 What was the total distance?

24 How long did the whole trip take?

25 When did the traveller stop?

26 In which hour was the speed greatest?

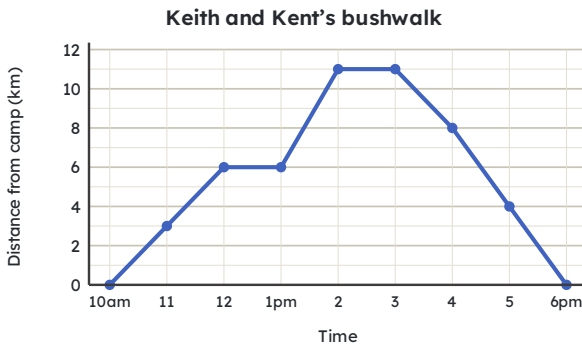
27 What was the average speed for the whole trip?

Level 2 · Building up

Medium · 1 question

WE DO With your teacher: copy what they write on the board.

Example 2 Use the travel graph.



a At what time did they start back?

Your teacher works this out. Copy every step.

b Between what times did they walk fastest?

Your teacher works this out. Copy every step.

YOU DO On your own: **Medium** Set C. Draw a travel graph for the journey.

1 QUESTION

REMEMBER A flat line means stopped.

1 Sam walks 4 km in 1 h, rests 30 min, walks 2 km in 30 min, then walks home (6 km) in 1.5 h. Start at 9 a.m.

Distance from home (km) against time

SKILL DRILL 2 Medium basics: keep practising

21 questions

YOU DO

On your own: **Medium** Skill drill 2. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 21 Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–15: find the missing value: $d = s \times t$, $s = d \div t$, $t = d \div s$.

1 $s = 40 \text{ km/h}$, $t = 4 \text{ h}$, $d = ?$

2 $d = 240 \text{ km}$, $s = 60 \text{ km/h}$, $t = ?$

3 $s = 50 \text{ km/h}$, $t = 5 \text{ h}$, $d = ?$

4 $d = 160 \text{ km}$, $t = 2 \text{ h}$, $s = ?$

5 $d = 160 \text{ km}$, $t = 4 \text{ h}$, $s = ?$

6 $d = 100 \text{ km}$, $t = 5 \text{ h}$, $s = ?$

7 $s = 50 \text{ km/h}$, $t = 3 \text{ h}$, $d = ?$

8 $d = 200 \text{ km}$, $s = 50 \text{ km/h}$, $t = ?$

9 $d = 80 \text{ km}$, $s = 20 \text{ km/h}$, $t = ?$

10 $s = 60 \text{ km/h}$, $t = 2 \text{ h}$, $d = ?$

11 $d = 80 \text{ km}$, $s = 40 \text{ km/h}$, $t = ?$

12 $d = 150 \text{ km}$, $s = 50 \text{ km/h}$, $t = ?$

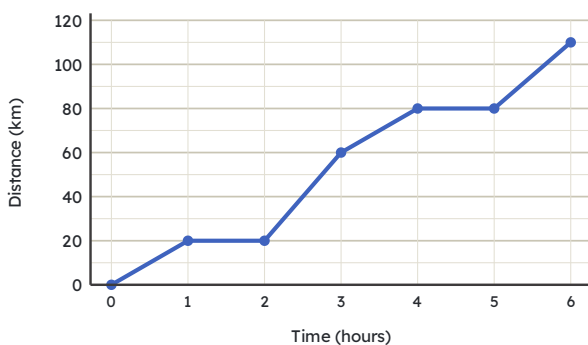
13 $s = 40 \text{ km/h}$, $t = 2 \text{ h}$, $d = ?$

14 $d = 150 \text{ km}$, $t = 3 \text{ h}$, $s = ?$

15 $s = 20 \text{ km/h}$, $t = 2 \text{ h}$, $d = ?$

ROUND 2 Questions 16–21: use the travel graph.

A bike ride



16 How far was travelled in the first hour?

17 What was the total distance?

18 How long did the whole trip take?

19 When did the traveller stop?

20 In which hour was the speed greatest?

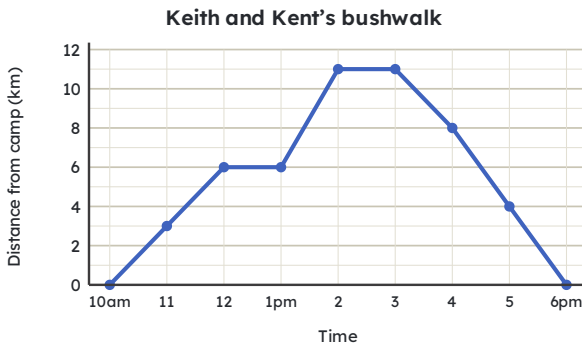
21 What was the average speed for the whole trip?

Level 3 · Challenge

Challenging · 4 questions

WE DO With your teacher: copy what they write on the board.

Example 3 Use the travel graph, then explain.



a Calculate their speed in the last 2 hours.

Your teacher works this out. Copy every step.

b What does a steeper line mean on a travel graph?

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Think it through.

4 QUESTIONS

REMEMBER Steeper line means faster.

1 On a travel graph, what does a horizontal line show?

2 What does a line going down to the time axis show?

3 Write a story for Mia's car trip in Set B.

4 Mia's average speed for the whole trip (including the stop)?

Extension · Stretch yourself

Finished Set D? Stretch yourself here.

Worked example

Two cyclists leave town A at 8 a.m. One rides at 20 km/h, the other at 25 km/h. How far apart are they at 10:30 a.m.?

1. Time = 2.5 h
2. Distances 50 km and 62.5 km

Answer: 12.5 km

YOU DO

On your own: read the worked example, then try these.

- E1 Draw both cyclists on one travel graph from 8 to 10:30 a.m.

- E2 When is the faster cyclist 10 km ahead?

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO

Show off your skill

Choose ONE question. Show your working.

Name _____ Class _____

Level 1

What does a flat section on a travel graph mean?

Level 2

A graph goes from 0 km at 1 p.m. to 30 km at 3 p.m. Find the speed.

Level 3

In Set A, how far from camp were they at 11 a.m.?

Lesson summary

Missed the lesson? Start here.

Remember

1. Time is across, distance is up.
3. Steeper line means faster.

2. A flat line means stopped.

4. Speed = change in distance ÷ change in time.

Level 1

Total walked

1. 11 km out, 11 km back

Answer: 22 km

Level 2

Stops

1. Two flat sections

Answer: 2

Level 3

Last 2 hours

1. $8 \text{ km} \div 2 \text{ h}$

Answer: 4 km/h

Extension

Cyclists at 10:30

1. $62.5 - 50$

Answer: 12.5 km

My reflection: what did I learn today? Which question was hardest, and why?

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
12.08

Time

Date _____

Learning intentions

We are learning to:

- round, convert, add and subtract time

Success criteria

I can:

- ☐ round time to the nearest hour or minute
- ☐ convert between hours, minutes and seconds
- ☐ add and subtract hours and minutes, carrying 60

Key terms Write the definition as your teacher explains it.

Minute

Second

Notes

Level 1 · Getting started

Easy · 18 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Round, then convert.

a Round 9 h 50 min to the nearest hour.

Your teacher works this out. Copy every step.

b Round 3.2 h to the nearest hour. How many minutes is 0.2 h?

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** Set A. Questions 1–6: Round to the nearest hour.

12 QUESTIONS

REMEMBER 60 s = 1 min, 60 min = 1 h, 24 h = 1 day.

1 9 h 50 min

2 3.2 h

3 4 h 12 min 49 s

4 7 h 30 min

5 1.75 h

6 12 h 29 min

Questions 7–12: convert.

7 2 h to min

8 90 min to h and min

9 3 min to s

10 150 s to min and s

11 0.5 h to min

12 1 day to h

SKILL DRILL 1 Easy basics: practise until it is automatic

36 questions

YOU DO

On your own: **Easy** Skill drill 1. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 36

Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–24: convert. 1 h = 60 min, 1 min = 60 s.

1 2 h to min

2 6 min to s

3 380 min to h and min

4 480 min to h

5 300 min to h

6 4 min to s

7 9 min to s

8 8 h to min

9 250 min to h and min

10 3 min to s

11 450 min to h and min

12 5 h to min

13 180 min to h

14 8 min to s

15 3 h to min

16 420 min to h

17 7 min to s

18 540 min to h

19 4 h to min

20 130 min to h and min

21 360 min to h

22 120 min to h

23 170 min to h and min

24 5 min to s

ROUND 2 Questions 25–36: add the times.

25 1 h 46 min + 1 h 16 min

26 1 h 4 min + 32 min

27 38 min + 1 h 18 min

28 42 min + 2 h 3 min

29 2 h 4 min + 2 h 5 min

30 22 min + 1 h 53 min

31 1 h 1 min + 1 h 24 min

32 2 h 54 min + 2 h 41 min

33 2 h 18 min + 1 h 43 min

34 2 h 33 min + 20 min

35 2 h 6 min + 26 min

36 2 h 35 min + 2 h 20 min

YOU DO

On your own: **Easy** Set B. Convert.

6 QUESTIONS

1 316 min to h and min

4 3 h 15 min to min

2 208 s to min and s

5 0.4 h to min

3 2.5 h to min

⑥ 1 day to min



Level 2 · Building up

Medium · 6 questions

WE DO

With your teacher: copy what they write on the board.

Example 2

Convert.

a Convert 316 minutes to hours and minutes.

Your teacher works this out. Copy every step.

b Convert 208 seconds to minutes and seconds.

Your teacher works this out. Copy every step.

YOU DO

On your own: **Medium** Set C. Evaluate.

6 QUESTIONS

REMEMBER Decimal hours: 0.1 h = 6 min.

① $6 \text{ h } 45 \text{ min} + 3 \text{ h } 20 \text{ min}$

[illegible]

3 $4 \text{ h } 33 \text{ min} - 2 \text{ h } 24 \text{ min}$

[illegible]

5 1 h 45 min $\times$ 4

[illegible]

② $3 \text{ h } 16 \text{ min} + 1 \text{ h } 26 \text{ min}$

Circumstance	Justified (%)	Not justified (%)
If someone is attacking you	85	15
If someone is threatening you	75	25
If someone is harassing you	65	35
If someone is insulting you	55	45
If someone is annoying you	15	85

④ $2\text{ h } 19\text{ min} - 1\text{ h } 50\text{ min}$

[illegible]

⑥ $7 \text{ h } 30 \text{ min} \div 5$

[illegible]

SKILL DRILL 2 Medium basics: keep practising

28 questions

YOU DO

On your own: **Medium** Skill drill 2. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 28

Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–12: subtract the times.

1 $5 \text{ h } 44 \text{ min} - 5 \text{ h } 13 \text{ min}$

2 $4 \text{ h } 27 \text{ min} - 1 \text{ h } 29 \text{ min}$

3 $2 \text{ h } 37 \text{ min} - 1 \text{ h } 46 \text{ min}$

4 $2 \text{ h } 41 \text{ min} - 54 \text{ min}$

5 $3 \text{ h } 30 \text{ min} - 1 \text{ h } 54 \text{ min}$

6 $2 \text{ h } 55 \text{ min} - 1 \text{ h } 12 \text{ min}$

7 $3 \text{ h } 46 \text{ min} - 3 \text{ h}$

8 $2 \text{ h } 44 \text{ min} - 2 \text{ h } 2 \text{ min}$

9 $5 \text{ h } 43 \text{ min} - 5 \text{ h } 25 \text{ min}$

10 $6 \text{ h } 5 \text{ min} - 4 \text{ h } 55 \text{ min}$

11 $4 \text{ h } 35 \text{ min} - 1 \text{ h } 33 \text{ min}$

12 $5 \text{ h } 34 \text{ min} - 3 \text{ h } 40 \text{ min}$

ROUND 2 Questions 13–28: change decimal hours to hours and minutes, or back.

13 $6 \text{ h } 24 \text{ min}$

14 4.2 h

15 6.75 h

16 5.5 h

17 6.4 h

18 3.2 h

19 $5 \text{ h } 12 \text{ min}$

20 6.5 h

21 $5 \text{ h } 24 \text{ min}$

22 $3 \text{ h } 6 \text{ min}$

23 $2 \text{ h } 30 \text{ min}$

24 $5 \text{ h } 30 \text{ min}$

25 $2 \text{ h } 24 \text{ min}$

26 $6 \text{ h } 15 \text{ min}$

27 $1 \text{ h } 30 \text{ min}$

28 4.5 h

Level 3 · Challenge

Challenging · 4 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Add and subtract times.

a $6 \text{ h } 45 \text{ min} + 3 \text{ h } 20 \text{ min}$

Your teacher works this out. Copy every step.

b $2 \text{ h } 19 \text{ min} - 1 \text{ h } 50 \text{ min}$

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Problems.

4 QUESTIONS

REMEMBER Add: add minutes, carry 60 min as 1 h.

- 1** A movie starts at 7:35 p.m. and runs for 2 h 18 min. When does it end?

- 3** A runner's lap times are 1 min 52 s and 2 min 07 s. Find the difference.

- 2** Three songs last 3 min 42 s, 4 min 15 s and 2 min 58 s. Find the total time.

- 4** Convert 2.85 h to hours and minutes.

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO **Show off your skill**

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Convert 150 min to hours and minutes.

Level 2

Evaluate $2 \text{ h } 40 \text{ min} + 1 \text{ h } 35 \text{ min}$.

Level 3

Evaluate $5 \text{ h } 10 \text{ min} - 2 \text{ h } 45 \text{ min}$.

 Extension · Stretch yourself Worked example

How many seconds are there in a week?

1. $60 \times 60 \times 24 \times 7$

Answer: 604 800 s

YOU DO

On your own: read the worked example, then try these.

E1 About how many hours old are you?

E2 Convert 1 000 000 seconds to days, hours, minutes and seconds.

Lesson summary

Missed the lesson? Start here.

Remember 1. $60 \text{ s} = 1 \text{ min}$, $60 \text{ min} = 1 \text{ h}$, $24 \text{ h} = 1 \text{ day}$.
3. Add: add minutes, carry 60 min as 1 h.

2. Decimal hours: $0.1 \text{ h} = 6 \text{ min}$.
4. Subtract: borrow 1 h as 60 min.

 Level 1

316 min

1. $316 \div 60 = 5 \text{ r } 16$

Answer: 5 h 16 min

 Level 2

6 h 45 + 3 h 20

1. 9 h 65 min

Answer: 10 h 5 min

 Level 3

2 h 19 – 1 h 50

1. 1 h 79 – 1 h 50

Answer: 29 min

 Extension

Seconds in a week

1. $60 \times 60 \times 24 \times 7$

Answer: 604 800

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
12.09

12-hour and 24-hour time

Date _____

Learning intentions

We are learning to:

- convert between 12-hour and 24-hour time

Success criteria

I can:

- ☐ use a.m. and p.m. correctly
- ☐ write 24-hour time with 4 digits
- ☐ convert midnight and noon times correctly

Key terms Write the definition as your teacher explains it.

a.m.

p.m.

24-hour time

Notes

Level 1 · Getting started

Easy · 18 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Convert to 12-hour time.

a Convert 15:50 to 12-hour time.

b Convert 01:48 to 12-hour time.

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** Set A. Questions 1–6: Convert to 12-hour time.

12 QUESTIONS

REMEMBER 24-hour time uses 4 digits: 00:00 to 23:59.

1 01:48

2 06:24

3 15:50

4 12:30

5 23:05

6 00:40

Questions 7–12: convert.

7 14:30 to 12-hour

8 09:15 to 12-hour

9 00:20 to 12-hour

10 7:45 p.m. to 24-hour

11 11:05 a.m. to 24-hour

12 12:10 a.m. to 24-hour

SKILL DRILL 1 Easy basics: practise until it is automatic

40 questions

YOU DO On your own: **Easy** Skill drill 1. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 40 Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–24: write in 24-hour time.

- | | | | |
|-----------------------|------------------------|-----------------------|------------------------|
| 1 11:05 p.m.
_____ | 2 7:45 p.m.
_____ | 3 7:15 p.m.
_____ | 4 8:05 a.m.
_____ |
| 5 12:50 p.m.
_____ | 6 2:10 p.m.
_____ | 7 3:50 a.m.
_____ | 8 7:25 a.m.
_____ |
| 9 9:00 a.m.
_____ | 10 4:35 a.m.
_____ | 11 3:00 a.m.
_____ | 12 2:25 a.m.
_____ |
| 13 2:40 p.m.
_____ | 14 12:50 a.m.
_____ | 15 4:50 a.m.
_____ | 16 12:20 a.m.
_____ |
| 17 1:05 a.m.
_____ | 18 11:20 a.m.
_____ | 19 4:25 a.m.
_____ | 20 9:55 p.m.
_____ |
| 21 1:15 a.m.
_____ | 22 9:15 p.m.
_____ | 23 7:30 a.m.
_____ | 24 8:25 p.m.
_____ |

ROUND 2 Questions 25–40: write in 12-hour time (a.m. or p.m.).

- | | | | |
|-------------------|-------------------|-------------------|-------------------|
| 25 19:15
_____ | 26 14:20
_____ | 27 05:55
_____ | 28 08:50
_____ |
| 29 08:45
_____ | 30 22:30
_____ | 31 12:10
_____ | 32 00:35
_____ |
| 33 18:50
_____ | 34 06:30
_____ | 35 19:20
_____ | 36 22:50
_____ |
| 37 03:45
_____ | 38 04:15
_____ | 39 03:20
_____ | 40 08:10
_____ |

YOU DO On your own: **Easy** Set B. Convert to 24-hour time.

6 QUESTIONS

1 2:10 p.m.

4 1:50 p.m.

2 5:33 a.m.

5 11:45 p.m.

3 12:19 a.m.

6 12:05 p.m.

Level 2 · Building up

Medium · 4 questions

WE DO With your teacher: copy what they write on the board.

Example 2 Convert between 12-hour and 24-hour time.

a Convert 12:30 to 12-hour time.

Your teacher works this out. Copy every step.

b Convert 2:10 p.m. to 24-hour time.

Your teacher works this out. Copy every step.

YOU DO On your own: **Medium** Set C. Use the bus timetable.

4 QUESTIONS

REMEMBER p.m. times (except 12) add 12 hours.

1 When does the first bus reach the Beach (12-hour time)?

2 How long does the trip from School to Beach take?

3 You must be at the Shops by 4 p.m. Which bus from School?

4 How often do the buses run?

Stop	Bus times (24-hour)		
School	15:10	15:40	16:10
Library	15:18	15:48	16:18
Shops	15:31	16:01	16:31
Beach	15:45	16:15	16:45

SKILL DRILL 2 Medium basics: keep practising

31 questions

YOU DO

On your own: **Medium** Skill drill 2. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 31 Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–16: convert. Take care near midnight and midday.

1 00:20

2 11:10 a.m.

3 11:15

4 12:25 a.m.

5 11:55 p.m.

6 12:45 a.m.

7 11:40

8 11:55 a.m.

9 00:30

10 12:30 a.m.

11 12:10

12 23:10

13 12:05 a.m.

14 11:25 a.m.

15 12:15 a.m.

16 11:00 p.m.

ROUND 2 Questions 17–31: what is the time after the minutes given? Use 24-hour time.

17 125 min after 06:05

18 10 min after 14:20

19 150 min after 23:50

20 50 min after 07:25

21 85 min after 06:15

22 110 min after 09:05

23 40 min after 19:10

24 130 min after 07:35

25 30 min after 20:45

26 35 min after 18:25

27 90 min after 07:20

28 30 min after 01:00

29 110 min after 01:05

30 140 min after 09:20

31 120 min after 03:50

Level 3 · Challenge

Challenging · 4 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Convert, then explain.

a Convert 12:19 a.m. to 24-hour time.

Your teacher works this out. Copy every step.

b Why do airlines and hospitals use 24-hour time?

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Problems.

4 QUESTIONS

REMEMBER 12:xx a.m. is 00:xx; 12:xx p.m. is 12:xx.

1 A flight leaves at 22:40 and lands 3 h 35 min later. Write the landing time in 24-hour and 12-hour time.

3 Write the time 20 minutes before midnight in 24-hour time.

2 A shift starts at 18:30 and ends at 02:15. How long is it?

4 Which is later: 7:15 p.m. or 18:45?

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO **Show off your skill**

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Convert 17:25 to 12-hour time.

Level 2

Convert 8:05 p.m. to 24-hour time.

Level 3

Convert 12:45 a.m. to 24-hour time.

 Extension · Stretch yourself Worked example

A clock shows 09:40. What time will it be 1000 minutes later?

1. 1000 min = 16 h 40 min
2. 09:40 + 16:40 = 26:20 → 02:20 the next day

Answer: 02:20 (2:20 a.m.)

YOU DO

On your own: read the worked example, then try these.

- E1** What time is 500 minutes after 21:15?

- E2** How many minutes are there from 07:35 to 16:10?

Lesson summary

Missed the lesson? Start here.

Remember

1. 24-hour time uses 4 digits: 00:00 to 23:59.
2. p.m. times (except 12) add 12 hours.
3. 12:xx a.m. is 00:xx; 12:xx p.m. is 12:xx.
4. Subtract 12 from 13:00 onwards for p.m.

 Level 1

15:50

1. $15 - 12 = 3$

Answer: 3:50 p.m.

 Level 2

01:48

1. Before 12, a.m.

Answer: 1:48 a.m.

 Level 3

2:10 p.m.

1. $2 + 12 = 14$

Answer: 14:10

 Extension

1000 min after 09:40

1. + 16 h 40 min

Answer: 02:20

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
12.10

Time differences

Date _____

Learning intentions

We are learning to:

- calculate time differences and ages

Success criteria

I can:

- ☐ count on from the earlier time to the next hour, then to the later time
- ☐ work across midnight
- ☐ calculate an age in years, months and days

Key terms Write the definition as your teacher explains it.

Elapsed time

Time difference

Notes

Level 1 • Getting started

Easy • 18 questions

WE DO With your teacher: copy what they write on the board.**Example 1** Find the time difference.**a** Find the time from 5:25 a.m. to 9:45 a.m.**b** Find the time from 11:55 p.m. to 7:30 a.m.

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** Set A. Questions 1–6: Find the time difference.

12 QUESTIONS

REMEMBER Count on to the next hour.**1** 5:25 a.m. to 9:45 a.m.**2** 1:15 p.m. to 8:10 p.m.**3** 07:50 to 14:20**4** 10:40 a.m. to 1:05 p.m.**5** 06:15 to 11:50**6** 9:35 a.m. to 12:00 noon

Questions 7–12: find the time difference.

7 2:00 p.m. to 5:30 p.m.**8** 8:45 a.m. to 10:00 a.m.**9** 09:20 to 13:05**10** 11:30 a.m. to 2:10 p.m.**11** 22:00 to 01:00**12** 6:15 a.m. to 6:50 a.m.

SKILL DRILL 1 Easy basics: practise until it is automatic

33 questions

YOU DO

On your own: **Easy** Skill drill 1. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 33

Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–18: find the time difference.

1 11:45 a.m. to 7:45 p.m.

2 1:20 p.m. to 6:05 p.m.

3 8:35 a.m. to 10:00 a.m.

4 7:55 a.m. to 2:00 p.m.

5 10:50 a.m. to 2:40 p.m.

6 10:05 a.m. to 5:50 p.m.

7 12:20 p.m. to 4:45 p.m.

8 8:50 a.m. to 1:25 p.m.

9 10:10 a.m. to 12:10 p.m.

10 7:05 a.m. to 12:30 p.m.

11 7:45 a.m. to 11:05 a.m.

12 10:35 a.m. to 3:55 p.m.

13 7:35 a.m. to 12:50 p.m.

14 8:25 a.m. to 2:40 p.m.

15 12:50 p.m. to 4:50 p.m.

16 8:00 a.m. to 1:25 p.m.

17 9:45 a.m. to 4:50 p.m.

18 7:45 a.m. to 10:30 a.m.

ROUND 2 Questions 19–33: find the time difference (24-hour time).

19 01:25 to 03:15

20 07:35 to 12:45

21 04:15 to 09:25

22 16:00 to 18:00

23 16:25 to 16:55

24 09:05 to 10:10

25 14:30 to 18:35

26 18:10 to 19:25

27 05:40 to 06:05

28 20:45 to 23:05

29 02:35 to 06:40

30 05:35 to 06:10

31 16:40 to 19:05

32 09:25 to 10:20

33 19:10 to 22:20

YOU DO

On your own: **Easy** Set B. Find the time difference (across midnight).

6 QUESTIONS

1 11:55 p.m. to 7:30 a.m.

4 10:20 p.m. to 6:05 a.m.

2 23:45 to 00:15

5 21:10 to 04:50

3 15:30 to 03:25 (next day)

6 8:30 p.m. to 2:15 a.m.



Level 2 · Building up

Medium · 6 questions

WE DO

With your teacher: copy what they write on the board.

Example 2

Find the time difference across midnight.

a Find the time from 07:50 to 14:20.

Your teacher works this out. Copy every step.

b Find the time from 23:45 to 00:15.

Your teacher works this out. Copy every step.

YOU DO

On your own: **Medium** Set C. Calculate.

6 QUESTIONS

REMEMBER Then count whole hours.

1 Days from 25 March to 12 April

3 Weeks and days from 1 May to 30 June

5 Your age in years, months and days today

2 Days from 15 January to 1 March (not a leap year)

4 Samantha: born 3 June 2004. Age on 16 August 2023?

6 Time from 8:45 a.m. Monday to 3:20 p.m. Tuesday

SKILL DRILL 2 Medium basics: keep practising

24 questions

YOU DO

On your own: **Medium** Skill drill 2. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 24 Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–12: find the time difference. It goes past midnight.

1 22:25 to 01:15

2 23:00 to 00:15

3 21:05 to 02:00

4 22:20 to 00:55

5 22:45 to 04:35

6 23:25 to 02:55

7 22:35 to 02:45

8 22:30 to 03:35

9 21:55 to 04:20

10 22:25 to 00:40

11 22:30 to 02:35

12 23:10 to 03:35

ROUND 2 Questions 13–24: time zones: Sydney is 2 hours ahead of Perth. Find the time.

13 12:00 p.m. in Perth. Sydney?

14 8:45 a.m. in Perth. Sydney?

15 3:45 p.m. in Sydney. Perth?

16 6:40 p.m. in Perth. Sydney?

17 7:55 a.m. in Perth. Sydney?

18 8:10 a.m. in Sydney. Perth?

19 11:05 a.m. in Perth. Sydney?

20 3:40 p.m. in Sydney. Perth?

21 7:15 a.m. in Perth. Sydney?

22 12:30 p.m. in Perth. Sydney?

23 7:40 a.m. in Sydney. Perth?

24 12:15 p.m. in Perth. Sydney?



Level 3 · Challenge

Challenging · 4 questions · then try the extension

WE DO With your teacher: copy what they write on the board.**Example 3** Find the age or the number of days.

- a** Samantha was born on 3 June 2004. Find her age on 16 August 2023.

Your teacher works this out. Copy every step.

- b** How many days from 25 March to 12 April?

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Problems.

4 QUESTIONS

REMEMBER Then the extra minutes. Add them up.

- 1** A flight leaves Sydney at 9:50 p.m. and arrives in Perth at 12:45 a.m. local time. Perth is 2 h behind Sydney. How long is the flight?

- 3** Jack is 12 years 7 months old. His sister is 3 years 9 months younger. How old is she?

- 2** A baby was born on 28 February 2024. How old is she (in days) on 1 April 2024?

- 4** A parking meter charges \$3.20 per hour or part hour. Find the cost from 10:35 a.m. to 1:50 p.m.

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO Show off your skillChoose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Find the time from 8:20 a.m. to 11:05 a.m.

Level 2

Find the time from 22:30 to 01:10.

Level 3

How many days from 20 May to 4 June?

 Extension · Stretch yourself Worked example

It is 10:00 a.m. in Sydney. What time is it in London (9 hours behind)?

1. $10:00 - 9 \text{ h} = 1:00$

Answer: 1:00 a.m.

YOU DO

On your own: read the worked example, then try these.

- E1 A video call is at 7:00 p.m. in Sydney. What time is it in London?

- E2 Tokyo is 1 hour behind Sydney. A plane leaves Tokyo at 20:30 local time and takes 9 h 30 min. When does it land (Sydney time)?

Lesson summary

Missed the lesson? Start here.

Remember

1. Count on to the next hour.
2. Then count whole hours.
3. Then the extra minutes. Add them up.
4. For ages: years, then months, then days.

 Level 1

5:25 to 9:45 a.m.

1. $35 \text{ min} + 3 \text{ h} + 45 \text{ min}$

Answer: 4 h 20 min

 Level 2

11:55 p.m. to 7:30 a.m.

1. $5 \text{ min} + 7 \text{ h } 30 \text{ min}$

Answer: 7 h 35 min

 Level 3

Samantha's age

1. $3 \text{ Jun } 2004 \rightarrow 3 \text{ Aug } 2023 \rightarrow 16 \text{ Aug}$

Answer: 19 y 2 m 13 d

 Extension

10 a.m. Sydney

1. $- 9 \text{ h}$

Answer: 1:00 a.m.

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
12.11

Timetables

Date _____

Learning intentions

We are learning to:

- read and use timetables

Success criteria

I can:

- ☐ read departure and arrival times
- ☐ find travel times by subtracting
- ☐ choose the right service to arrive on time

Key terms Write the definition as your teacher explains it.

Timetable

Departure

Notes

Level 1 · Getting started

Easy · 16 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Use the train timetable.

Station	a.m.	a.m.	a.m.	p.m.
Moss Vale	6:05	7:40	9:10	7:45
Bowral	6:12	7:47	9:17	7:52
Mittagong	6:20	7:55	9:25	8:00
Picton	6:58	8:33	10:03	8:38
Campbelltown	7:25	9:00	10:30	9:05
Sydney (Central)	8:21	9:56	11:26	10:01

a When does the 7:40 a.m. train reach Picton?

Your teacher works this out. Copy every step.

b How long does the 7:45 p.m. train take to get to Sydney?

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** Set A. Use the train timetable.

4 QUESTIONS

REMEMBER Find the row for the station and the column for the service.

Station	a.m.	a.m.	a.m.	p.m.
Moss Vale	6:05	7:40	9:10	7:45
Bowral	6:12	7:47	9:17	7:52
Mittagong	6:20	7:55	9:25	8:00
Picton	6:58	8:33	10:03	8:38
Campbelltown	7:25	9:00	10:30	9:05
Sydney (Central)	8:21	9:56	11:26	10:01

① When does the 6:05 a.m. train reach Sydney?

② How long does it take?

③ When does the 9:10 a.m. train leave Mittagong?

④ How long from Picton to Sydney?

YOU DO On your own: **Easy** Set A2. Quick drill: how long is each trip?

6 QUESTIONS

① 8:10 → 8:55

② 9:40 → 11:05

③ 13:15 → 14:50

④ 6:05 → 8:21

⑤ 15:45 → 16:20

⑥ 7:52 → 10:01

YOU DO On your own: **Easy** Set B. Use the train timetable in Set A.

6 QUESTIONS

① When does the 7:40 a.m. train reach Picton?

④ Bowral to Campbelltown on the 9:10 a.m. train?

② How long does the 7:45 p.m. train take to reach Sydney?

⑤ Write the 7:45 p.m. arrival in 24-hour time.

③ Latest train from Moss Vale to be in Sydney by 11 a.m.?

⑥ How long between the first and second trains?

SKILL DRILL 1 Easy basics: practise until it is automatic

24 questions

YOU DO On your own: **Easy** Skill drill 1. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 24 Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–8: use the bus timetable.

Stop	Bus 1	Bus 2	Bus 3	Bus 4
School	08:00	08:35	09:00	09:50
Library	08:07	08:42	09:07	09:57
Pool	08:19	08:54	09:19	10:09
Shops	08:28	09:03	09:28	10:18
Beach	08:42	09:17	09:42	10:32

- 1 When does Bus 1 reach the Pool? _____
- 2 When does Bus 2 reach the Shops? _____

- 3 When does Bus 3 reach the Beach? _____
- 4 When does Bus 4 reach the Library? _____

- 5 How long from School to Beach? _____
- 6 How long from Library to Shops? _____

- 7 How long from Pool to Beach? _____
- 8 How long from School to Pool? _____

ROUND 2 Questions 9–24: how long is each trip?

- 9 11:15 → 13:20 _____
- 10 09:30 → 12:50 _____
- 11 08:55 → 10:05 _____
- 12 11:40 → 11:55 _____

- 13 08:25 → 09:25 _____
- 14 13:25 → 15:10 _____
- 15 16:30 → 17:05 _____
- 16 15:50 → 16:40 _____

- 17 10:05 → 12:35 _____
- 18 08:10 → 10:50 _____
- 19 15:15 → 16:15 _____
- 20 14:45 → 16:00 _____

- 21 12:20 → 13:10 _____
- 22 15:40 → 16:15 _____
- 23 11:50 → 13:45 _____
- 24 08:50 → 10:55 _____

Level 2 · Building up

Medium · 4 questions

WE DO With your teacher: copy what they write on the board.**Example 2** Use the train timetable.

Station	a.m.	a.m.	a.m.	p.m.
Moss Vale	6:05	7:40	9:10	7:45
Bowral	6:12	7:47	9:17	7:52
Mittagong	6:20	7:55	9:25	8:00
Picton	6:58	8:33	10:03	8:38
Campbelltown	7:25	9:00	10:30	9:05
Sydney (Central)	8:21	9:56	11:26	10:01

- a** What is the latest train you can catch at Moss Vale to be in Sydney by 11 a.m.?

Your teacher works this out. Copy every step.

- b** How long is the trip from Bowral to Campbelltown on the 9:10 a.m. train?

Your teacher works this out. Copy every step.

YOU DO On your own: **Medium** Set C. Use the bus timetable.

4 QUESTIONS

REMEMBER Read across or down carefully.

Stop	Bus times (24-hour)		
School	15:10	15:40	16:10
Library	15:18	15:48	16:18
Shops	15:31	16:01	16:31
Beach	15:45	16:15	16:45

- 1** How long does it take from Library to Beach?

- 2** You arrive at the Library at 15:50. When is the next bus? When will you reach the Beach?

- 3** You need to be at the Shops by 4:30 p.m. What is the latest bus from School?

- 4** Write the times for the 16:10 bus in 12-hour time.

SKILL DRILL 2 Medium basics: keep practising

21 questions

YOU DO

On your own: **Medium** Skill drill 2. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 21 Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–6: use the bus timetable.

Stop	Bus 1	Bus 2	Bus 3	Bus 4
School	08:00	08:35	09:00	09:50
Library	08:07	08:42	09:07	09:57
Pool	08:19	08:54	09:19	10:09
Shops	08:28	09:03	09:28	10:18
Beach	08:42	09:17	09:42	10:32

- 1 You must be at the Beach by 09:47. Which is the latest bus from School?

- 2 You reach the Library at 08:44. When is the next bus?

- 3 Which bus takes the longest from School to Beach?

- 4 How long do you wait at the Pool if you arrive at 08:50?

- 5 Write Bus 4's time at the Shops in 12-hour time.

- 6 How long is the whole route?

ROUND 2 Questions 7–21: how long is each trip? It goes past the hour.

- 7 8:50 a.m. → 11:05 a.m.

- 8 10:45 a.m. → 1:10 p.m.

- 9 2:40 p.m. → 4:35 p.m.

- 10 10:50 a.m. → 1:20 p.m.

- 11 11:55 a.m. → 12:15 p.m.

- 12 7:55 a.m. → 8:50 a.m.

- 13 5:50 p.m. → 7:25 p.m.

- 14 10:45 a.m. → 11:45 a.m.

- 15 1:45 p.m. → 3:50 p.m.

- 16 8:55 a.m. → 10:25 a.m.

- 17 12:40 p.m. → 3:05 p.m.

- 18 11:45 a.m. → 12:45 p.m.

- 19 12:50 p.m. → 1:40 p.m.

- 20 11:55 a.m. → 12:25 p.m.

- 21 3:55 p.m. → 6:00 p.m.

Level 3 · Challenge

Challenging · 2 questions

WE DO With your teacher: copy what they write on the board.

Example 3 Use the train timetable, then explain.

Station	a.m.	a.m.	a.m.	p.m.
Moss Vale	6:05	7:40	9:10	7:45
Bowral	6:12	7:47	9:17	7:52
Mittagong	6:20	7:55	9:25	8:00
Picton	6:58	8:33	10:03	8:38
Campbelltown	7:25	9:00	10:30	9:05
Sydney (Central)	8:21	9:56	11:26	10:01

a Which train arrives in Sydney closest to 10 a.m.?

Your teacher works this out. Copy every step.

b Why are timetables often written in 24-hour time?

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Make your own timetable.

2 QUESTIONS

REMEMBER Travel time = arrival – departure.

1 A ferry leaves the wharf every 40 minutes from 06:50 until 09:30. The trip takes 25 minutes. Write a timetable showing departure and arrival times.

Departs	Arrives

2 Your school day as a timetable: list the start and end of each period in 24-hour time.



Extension · Stretch yourself

Finished Set D? Stretch yourself here.



Worked example

You live 10 minutes' walk from Bowral station. You must be in Sydney by 10:15 a.m. What time must you leave home?

1. The 7:40 a.m. train arrives at 9:56 a.m.
2. It leaves Bowral at 7:47 a.m.
3. Leave home 10 min earlier

Answer: 7:37 a.m. (or earlier)

YOU DO

On your own: read the worked example, then try these.

- E1** Your friend at Picton wants to meet you on the same train. Which time does she board?

- E2** The 7:40 a.m. train is running 12 minutes late. When will it reach Sydney?

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO

Show off your skill

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

When does the 9:10 a.m. train reach Campbelltown?

Level 2

How long is the bus trip from School to Shops?

Level 3

Which train should you catch from Moss Vale to reach Sydney before 9 a.m.?

Lesson summary

Missed the lesson? Start here.

Remember

1. Find the row for the station and the column for the service.
2. Read across or down carefully.
3. Travel time = arrival – departure.
4. Choose the service that arrives before you need to be there.

Level 1

7:45 p.m. to Sydney

1. 7:45 p.m. → 10:01 p.m.

Answer: 2 h 16 min

Level 2

In Sydney by 11 a.m.

1. 9:10 arrives 11:26 (too late)

Answer: 7:40 a.m.

Level 3

Library to Beach

1. 15:18 → 15:45

Answer: 27 min

Extension

Leave home for Bowral

1. 7:47 – 10 min

Answer: 7:37 a.m.

My reflection: what did I learn today? Which question was hardest, and why?

This space is left blank on purpose. It is the back of your exit ticket.